

BILIARTO SASTRO CEMERSON

Yogyakarta, Indonesia
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Data Engineer with hands-on experience building end-to-end **data pipelines** and analytics-ready data models. Skilled in designing **ETL/ELT** workflows, medallion architecture, and **data quality** checks using tools such as **Python, SQL, DuckDB, dbt, and Airflow**, with a strong analytical foundation from data science projects and competitions.

EDUCATION

Bachelor of Science in Statistics | Yogyakarta State University 2021 - 2025

- GPA, 3.65/4.00 ([credits](#))
- Actively led and contributed to multiple data analysis projects and competitions.

EXPERIENCE

Data Annotator & AI Reviewer Aug 2025 - Present
Simplifa.AI | Yogyakarta, Indonesia

- Validated and standardized **1,000+ bank transaction records** across 20+ formats, ensuring high-quality structured data for downstream processing.
- Labeled and curated **5,000+ financial data points** using Argila and HuggingFace, contributing to dataset quality and consistency for model training.

Data Analyst Intern Jul - Aug 2023
Rumah Sakit Pratama Kota Yogyakarta | Yogyakarta, Indonesia

- Analyzed medical records of **500+ COVID-19 patients**, uncovering symptom patterns and risk factors.
- Built classification models (**Binary Logistic Regression**) to support clinical risk stratification .

PROJECTS

BMKG Earthquake Monitoring Pipeline ([Repo](#))

- Built a production-style batch pipeline ingesting the **15 most recent felt-earthquake events** from the BMKG API into a bronze layer (raw JSON snapshots) for auditability and replay. Transformed data through incremental dbt models with deduplication via stable event IDs, magnitude/depth categorization, and data validity checks, enforcing freshness SLAs via automated CI with GitHub Actions to ensure idempotent, observable, and backfillable delivery.
- Tech Stack:** Python, SQL, DuckDB, dbt, GitHub Actions

Indonesia Geopolitik Risk (GPR) Index ([Repo](#))

- Engineered a zero-cost automated ELT pipeline replicating the Caldara & Iacoviello GPR Index methodology for Indonesia, processing an **average of 350+ news articles per day** from major national outlets. dbt models aggregate daily keyword frequencies into 5 geopolitical sub-indexes (war threat, peace threat, military buildup, war acts, terror acts), producing a composite GPR score as an actionable economic risk indicator.
- Tech Stack:** Python, SQL, DuckDB, dbt, GitHub Actions

CERTIFICATION

DeepLearning.AI Data Engineering by DeepLearning.AI & AWS on Coursera ([cert](#)) Jul 2025
Python Project for Data Engineering by IBM on Coursera ([cert](#)) Jun 2025
Google Cloud Fundamentals: Core Infrastructure by Google Cloud on Coursera ([cert](#)) Apr 2025

HONORS AND AWARDS

2nd Winner - DATAVERS ANAVA UGM 2025 ([cert](#)) Jan – Feb 2025
Finalist (Top 12) - Big Data Challenge (BDC) Satria Data 2024 ([cert](#)) Jun – Aug 2024
3rd Winner - Olimpiade Nasional Statistika (ONS) ANAVA UGM 2024 ([cert](#)) Jan – Feb 2024
Finalist (Top 5) - Turnamen Sains Data Nasional (TSDN) 2023 ([cert](#)) Oct – Dec 2023

ADDITIONAL INFORMATION

Skills: ETL / ELT, Batch Processing, Data Warehousing, Data Modeling, Data Quality, Big Data.
Tools: Python, SQL, DuckDB, dbt, Apache Airflow, Apache Spark, Git/GitHub, Cloud (AWS/GCP).
Languages: Indonesian (Native), English (Intermediate).